A Rotation Diagnostic for Proposition 1

in Daniels–Rigollet's *Splat Regression Models*

Feier Lyu

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Purpose. While reading Daniels and Rigollet's *Splat Regression Models* [1], I used a simple rotation test to understand the geometric interpretation of Proposition 1. The question is whether centered isotropy alone makes the stated Bures expression an ambient Wasserstein identity, or whether an additional compatibility assumption or a covariance-quotient interpretation is intended.

The paper considers affine splats

$$\rho_{A,b} := (A(\cdot) + b)_{\#}\rho, \qquad A \in \mathbb{R}^{d\times d}, \quad b \in \mathbb{R}^d,$$

where the mother splat ρ is centered and isotropic. Proposition 1 states that

$$W_2^2(\rho_{A,b}, \rho_{R,s}) = \|b - s\|_2^2 + \|A\|_F^2 + \|R\|_F^2 - 2\left\|A^T R\right\|_* . \tag{1}$$

The right-hand side is the squared Bures–Gelbrich expression for the corresponding means and covariances. It equals the squared ambient W_2 distance for Gaussian measures; for general measures with finite second moments, it lower-bounds the squared ambient distance by Gelbrich's inequality [2]. The issue is whether a common centered isotropic mother splat makes this bound exact throughout its affine orbit.

A rotation diagnostic

Take

$$b = s = 0, \qquad A = \mathrm{Id}, \qquad R = Q,$$

where $Q \in O(d)$. The right-hand side of (1) becomes

$$\|\mathrm{Id}\|_F^2 + \|Q\|_F^2 - 2\|Q\|_* = d + d - 2d = 0.$$

If (1) is interpreted as the ambient W_2 distance between the full pushforward measures, then

$$W_2(\rho, Q_{\#}\rho) = 0,$$

and hence

$$\rho = Q_{\#}\rho. \tag{2}$$

Thus, validity for every orthogonal Q forces the mother splat to be spherically symmetric. Centered isotropy only gives

$$\mathbb{E}X = 0, \qquad \mathrm{Cov}(X) = I_d,$$

and does not imply rotational invariance.

For a concrete example, let $d = 2$ and

$$X = (U, V), \qquad U, V \text{ i.i.d.} \sim \mathrm{Unif}[-\sqrt{3}, \sqrt{3}],$$

so that $\mathbb{E}X = 0$ and $\mathrm{Cov}(X) = I_2$. Consider the $45°$ rotation

$$Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

If $\rho = \mathcal{L}(X)$ were equal to $Q_{\#}\rho$, then U and $(U - V)/\sqrt{2}$ would have the same fourth moment. But

$$\mathbb{E}[U^4] = \frac{9}{5},$$

while independence and $\mathbb{E}[U^2] = \mathbb{E}[V^2] = 1$ give

$$\begin{aligned} \mathbb{E}\left[\left(\frac{U - V}{\sqrt{2}}\right)^4\right] &= \frac{1}{4}\left(2\mathbb{E}[U^4] + 6\mathbb{E}[U^2]\mathbb{E}[V^2]\right) \\ &= \frac{12}{5}. \end{aligned}$$

Therefore $\rho \neq Q_{\#}\rho$, and consequently

$$W_2(\rho, Q_{\#}\rho) > 0,$$

even though the right-hand side of (1) is zero.

Geometric interpretation

If ρ is not spherically symmetric, $A_{\#}\rho$ and $(AQ)_{\#}\rho$ can be different measures, while their covariance matrices agree:

$$(AQ)(AQ)^T = AA^T.$$

The Bures expression therefore assigns zero distance to A and AQ. This is natural on the covariance quotient

$$A \sim AQ, \qquad Q \in O(d),$$

but not, in general, as the restriction of ambient W_2 to the full affine-pushforward family. Before quotienting, the expression is a pseudometric on the matrix factors.

The same distinction appears in an optimal-map argument. Convex-gradient form establishes optimality only after the map is shown to push the specified source to the specified target. In the rotation test, that identity is exactly (2); matching the first two moments does not imply it.

Is Proposition 1 intended to assume spherical symmetry, or a more general condition ensuring that the covariance-level affine map transports the full distributions? Alternatively, is the Bures expression intended as a metric on the covariance quotient rather than as ambient W_2 between the full pushforward measures?

This also affects how I read the later gradient-flow calculations: is the splat family an isometrically embedded subset of $\mathcal{P}_2(\mathbb{R}^d)$, or a parameter space endowed with a tractable Bures-type computational geometry?

References

[1] M. Daniels and P. Rigollet. *Splat Regression Models*. arXiv:2511.14042, 2025.

[2] M. Gelbrich. On a formula for the L^2-Wasserstein metric between measures on Euclidean and Hilbert spaces. *Mathematische Nachrichten*, 147:185–203, 1990. doi:10.1002/mana.19901470121.